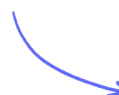


Proportion

Direct Proportion / Inverse Proportion / Variation Models

Medium (4 questions)	/11
Hard (4 questions)	/9
Very Hard (5 questions)	/15
Total Marks	/35

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Medium Questions

- 1 The following table of data represents the length, x and area, y of some irregular shapes.

x	1	5	b
y	a	5	20

Given that y is proportional to the square of x , find the values of a and b .

(2 marks)

2 $y \propto \frac{1}{\sqrt{x}}$

When $y = 8$, $x = 4$.

Find y when $x = 49$.

$y = \dots\dots\dots$

(3 marks)

- 3 y varies directly with the square root of x .

When $x = 9$, $y = 6$.

Find y when $x = 25$.

$y = \dots\dots\dots$

(3 marks)

- 4 y is inversely proportional to x .
When $x = 9$, $y = 8$.

Find y when $x = 6$.

$y = \dots\dots\dots$

(3 marks)

Hard Questions

- 1 y varies inversely as the square root of $(x + 1)$.

When $x = 8$, $y = 5$.

Find y in terms of x .

(2 marks)

- 2 y is directly proportional to $(x - 4)$.

When $x = 16$, $y = 3$.

Find y in terms of x .

$y =$

(2 marks)

- 3 y is inversely proportional to x^3 .

When $x = 2$, $y = 0.5$.

Find y in terms of x .

$y =$

(2 marks)

- 4 y is directly proportional to $(x - 1)^2$.

When $x = 3$, $y = 24$.

Find y when $x = 6$.

$y = \dots\dots\dots$

(3 marks)

Very Hard Questions

- 1 (a)** The table below shows the time, t hours, it takes a car to travel a certain distance at different speeds, s miles per hour.

s	30	40	50	60
t	4	3	a	2

State what type of variation model would be suitable for this data.

(1 mark)

- (b)** Find an equation for t in terms of s .

(2 marks)

- (c)** Find the value of a .

(1 mark)

- (d)** State the total distance that the car has driven.

(1 mark)

- (e)** Use the model to calculate the speed if a car takes 30 minutes to complete the journey.

(1 mark)

- (f)** Give a reason why using the model to answer part (e) is not suitable.

(1 mark)

- 2** m varies inversely with the square of $(p - 1)$.

When $p = 4$, $m = 5$.

Find m when $p = 6$.

$$m = \dots\dots\dots$$

(2 marks)

- 3** y is inversely proportional to the square root of $(x + 1)$.

When $x = 8$, $y = 2$.

Find y when $x = 99$.

$$y = \dots\dots\dots$$

(3 marks)

- 4** y is inversely proportional to $(x + 3)^2$

When $x = 2$, $y = 8$.

Find y when $x = 7$.

$$y = \dots\dots\dots$$

(1 mark)

- 5** y varies inversely with the square of $(x + 2)$.

When $x = 3$, $y = 2$.

Find y when $x = 8$.

$$y = \dots\dots\dots$$

(2 marks)